

OSL-SAR dating and technological characterization of archaeological ceramics from the Tupiguarani ceramic tradition groups: Pirajuí I and Arataca II, Pernambuco, Brazil

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1. INTRODUCTION

In this study, we present chronological and archaeometric results from clay vessel fragments of the Tupiguarani ceramic tradition, recovered from archaeological contact sites in northeastern Brazil (Figure 1). In Pernambuco, the Tupiguarani tradition is recognized by its characteristic ceramics, shaped by processes of migration and territorial expansion [3]. Our aim was to characterize the manufacturing technology of these vessels through chronological, mineralogical, chemical, and firing analyses, in order to understand the technical and cultural choices along the chaîne opératoire [4].

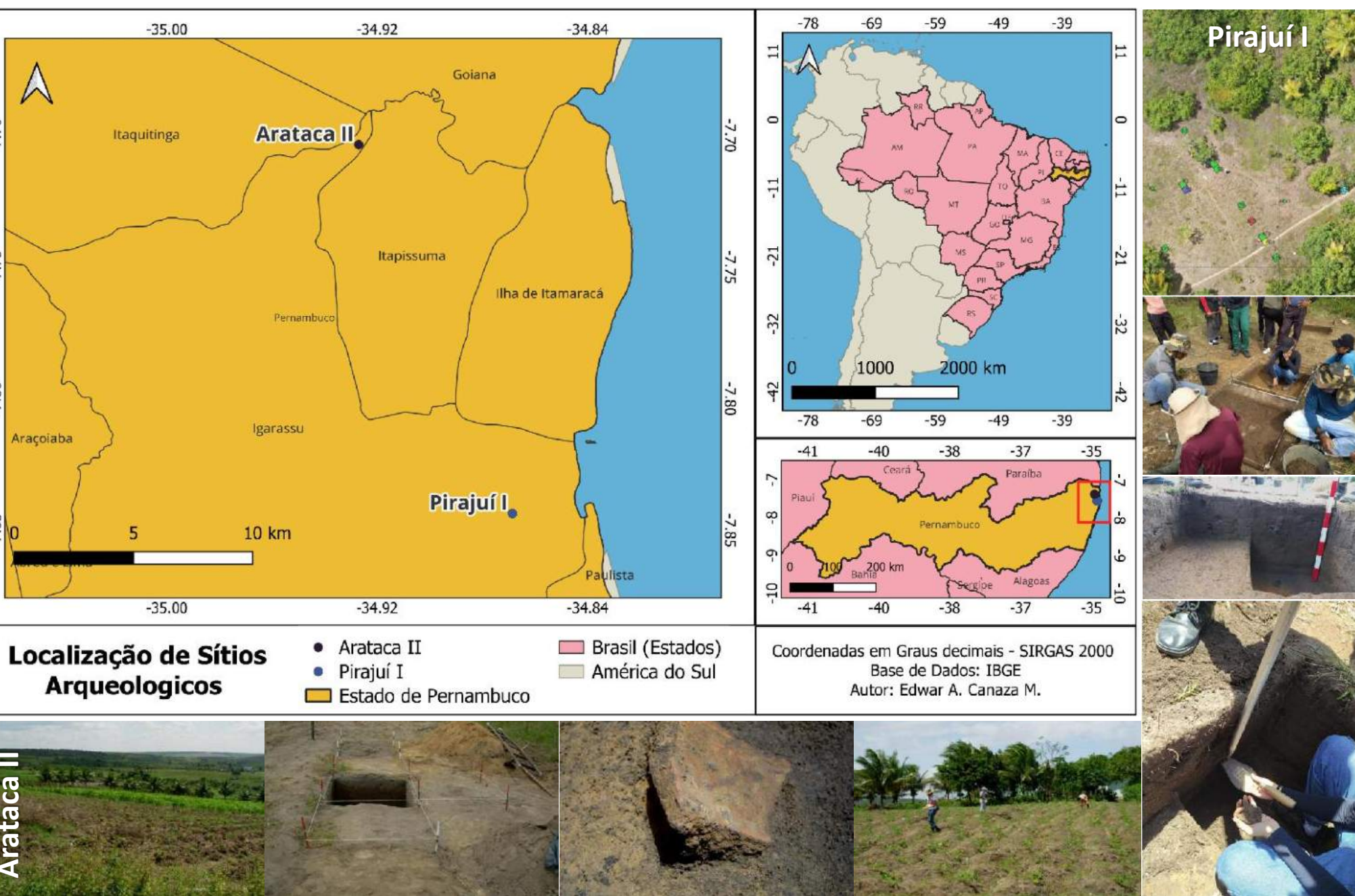


Figure 1. Location map of the archaeological sites in Pernambuco, Brazil

2. MATERIALS AND METHODS

We analyzed pottery vessel fragments, mainly rims (five per site). OSL dating was performed on pure quartz grains (75–180 µm) after standard pretreatment [1], with SAR analyses carried out using the Luminescence package in R [2] (Figure 2). The dose rate was determined via high-resolution gamma spectrometry with a high-purity germanium detector (Canberra system), measuring U, Th, and K contents in the pottery and surrounding sediments, and also considering theoretical values of cosmic radiation [1]. Mineralogical and firing analyses were conducted on <45 µm powdered cores, while 30 µm thin sections were prepared for petrography [5]. Additional measurements included XRD (Rigaku MiniFlex 600), XRF (Rigaku Supermini200, WDXRF), EPR (Magnetech ESR5000, X-band), and petrographic observations under PPL/XPL (Olympus BX51).



3. RESULTS

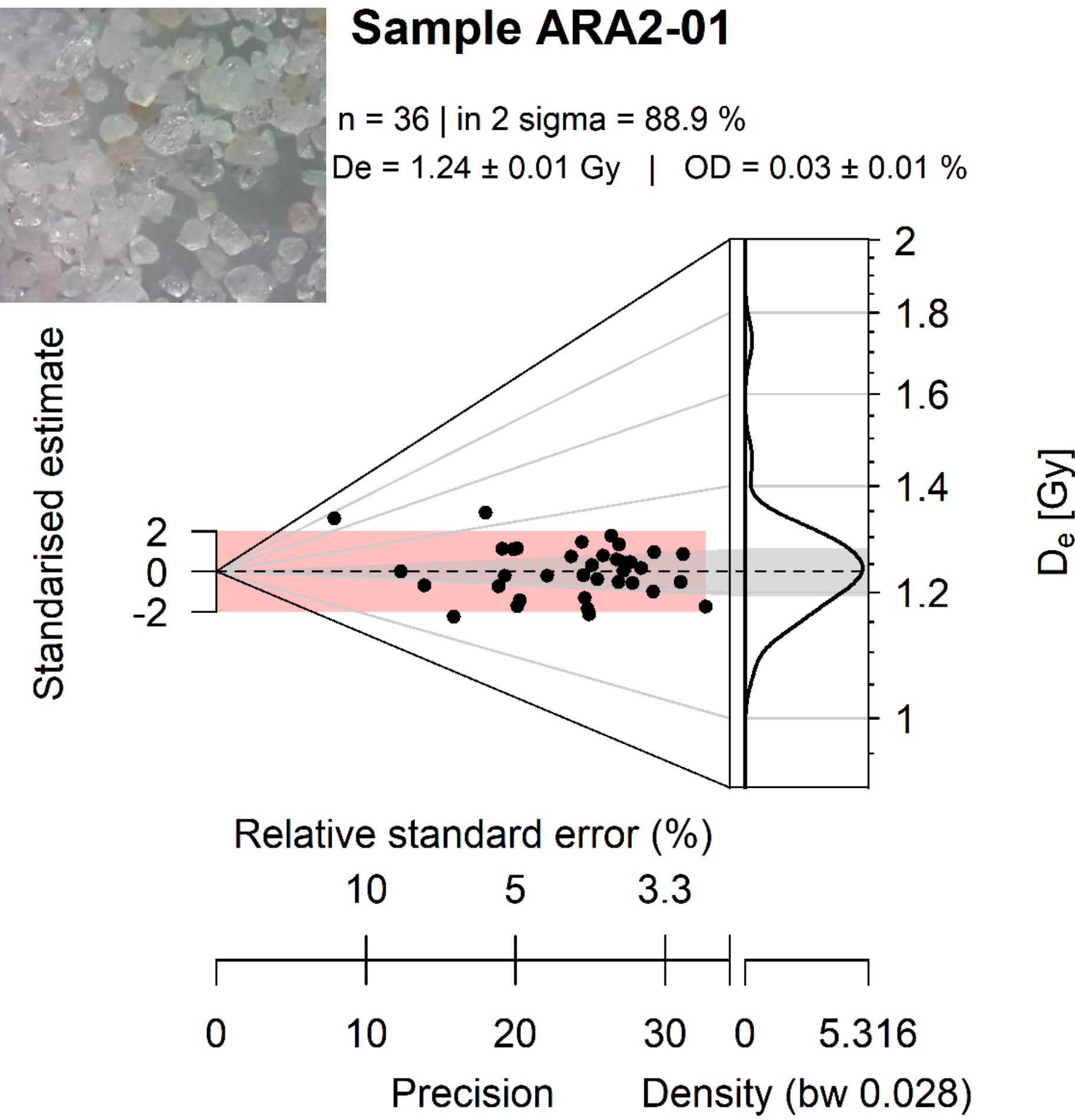


Figure 2. Radial plot of equivalent dose (D_e) values obtained using the OSL-SAR protocol for sample ARA2-01. Each symbol corresponds to one aliquot.

Table 1. Equivalent dose, annual dose rate, and calculated OSL ages

Sample	n Aliquot	D_e (Gy)	TDa (mGy/years)	Age (years)
ARA2-01	36	1.237 ± 0.018	1.203 ± 0.026	1028 ± 27
ARA2-02	25	1.375 ± 0.036	1.217 ± 0.027	1130 ± 39
PIRA1-01	45	1.764 ± 0.033	1.210 ± 0.025	1458 ± 41
PIRA1-02	46	1.594 ± 0.028	1.210 ± 0.025	1317 ± 36

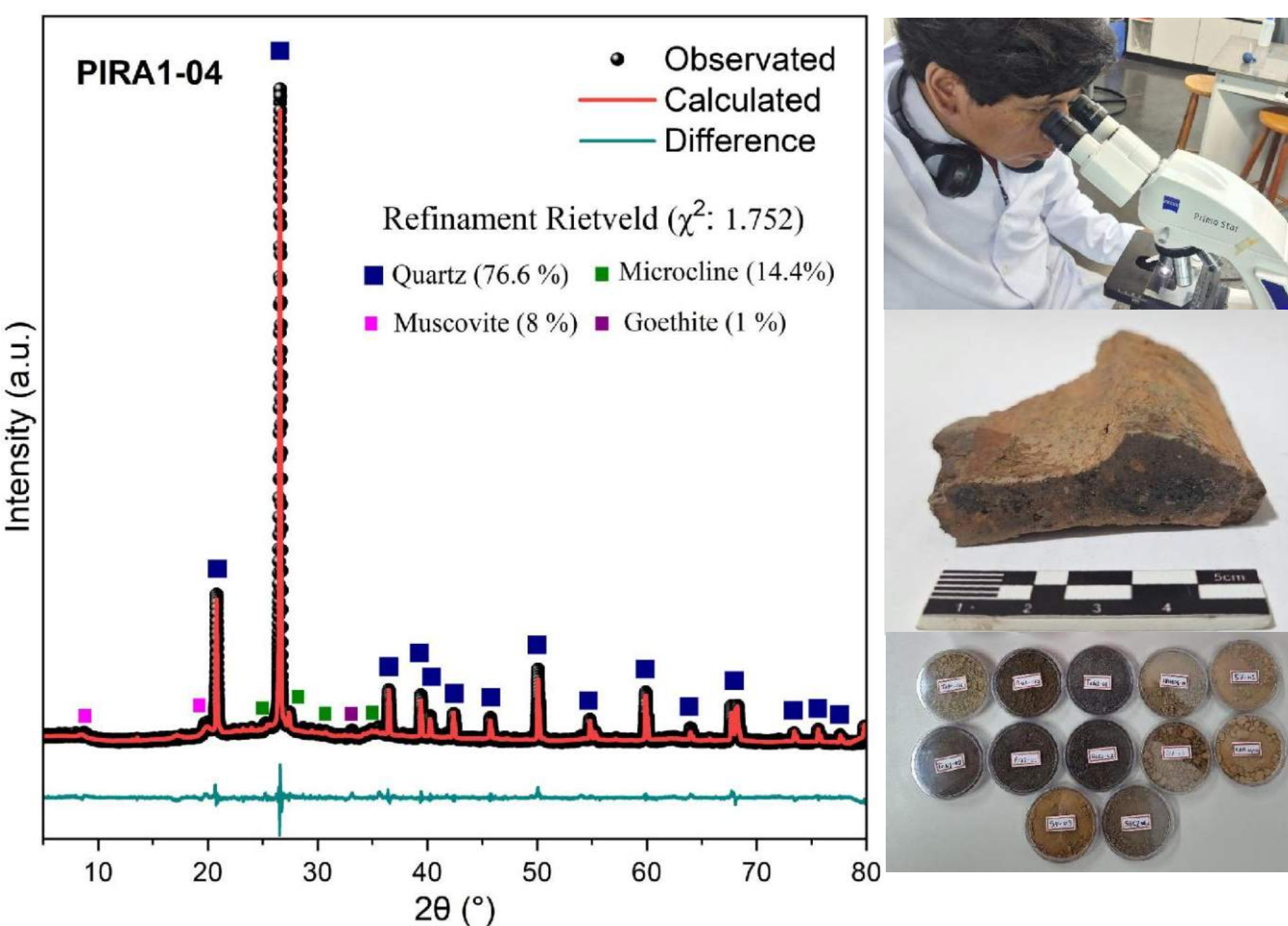


Figure 3. XRD pattern with identified mineral phases and their relative abundances obtained by Rietveld refinement

Table 2. Mineral phases identified by Rietveld refinement and their abbreviations (Q: Quartz; Mi: Microcline; Mu: Muscovite; Go: Goethite)

Sample	Minerals phases (%)				χ^2
	Q	Mi	Mu	Go	
ARA2-02	73.8	4.4	21.8	-	1.69
ARA2-03	65.4	13.3	21.3	-	1.63
ARA2-04	77.8	9.4	12.9	-	1.61
PIRA1-01	90.5	4.1	5.3	-	1.86
PIRA1-04	76.6	14.4	8	1	1.75
PIRA1-05	75.2	0.3	24.1	0.5	1.29

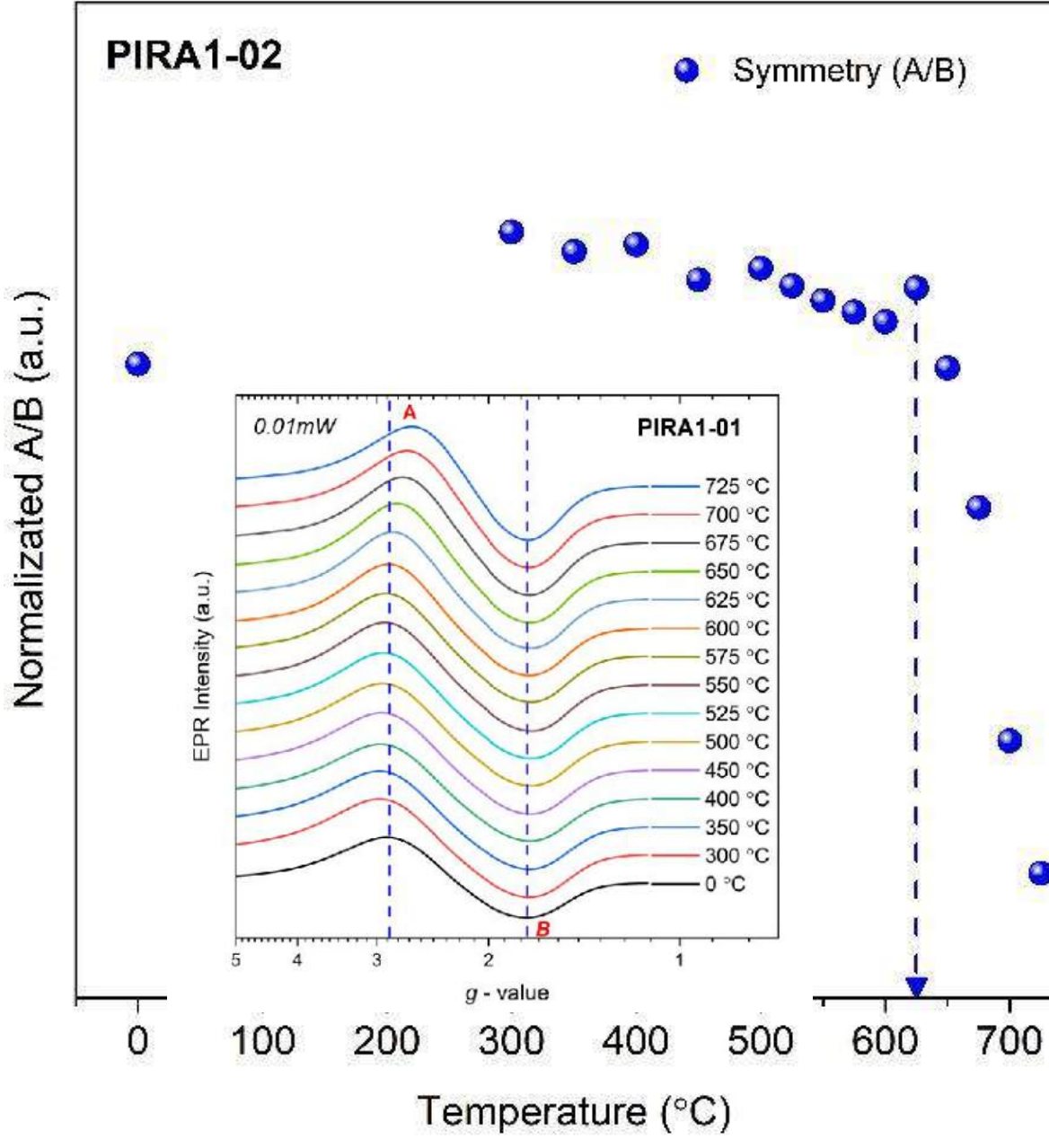


Figure 4. Symmetry parameter A/B of the Fe EPR signals in clay measured at different temperatures, with A and B as the signal maximum and minimum. Inset shows all individual measurements

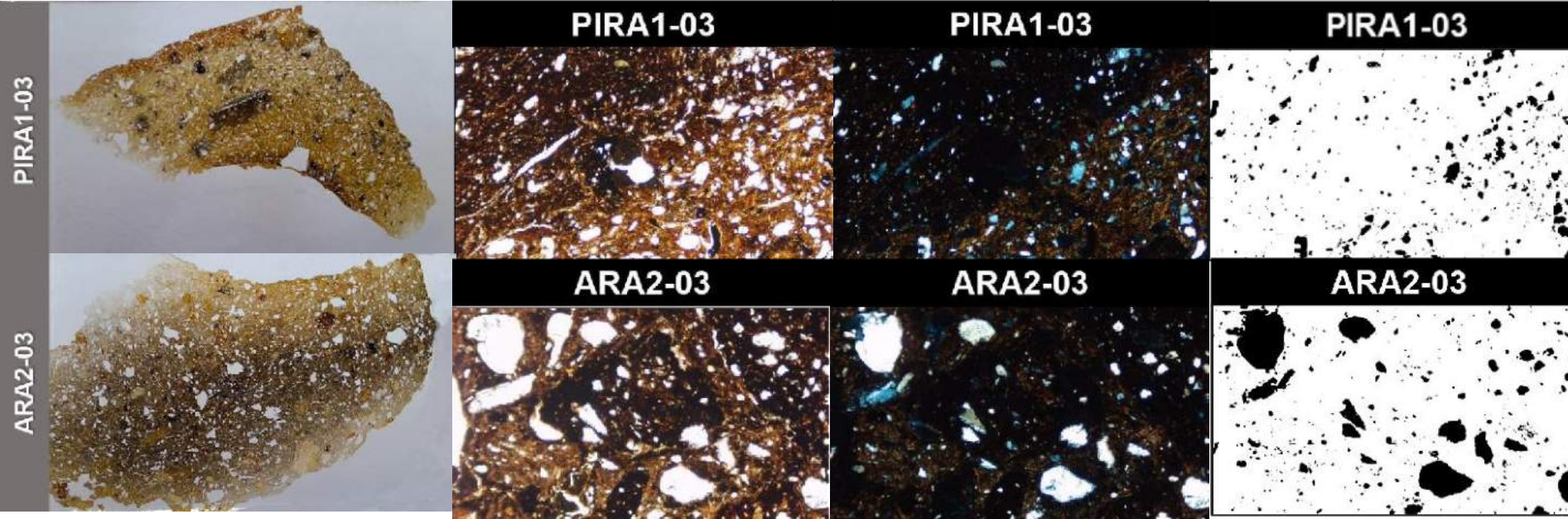


Figure 5. Petrography of a ceramic fragment showing quartz grains in PPL, XPL, and processed with ImageJ to highlight grains and calculate relative area.

Table 4. Percentage of quartz grains and area occupied in thin-section microphotographs analyzed with ImageJ

Sample	N - Grains (%) / Area (%)				N Total / (4.55 mm ²)
	Silt	Fine	Medium	Coarse	
ARA2-02	93.6/1.3	5.5/1.0	0.9/2.6	-	329
ARA2-03	90.6/1.8	7.6/2.8	1.8/8.8	-	445
PIRA1-01	80.3/0.8	14.8/5.1	4.5/12.0	0.41/12.7	243
PIRA1-03	91.3/2.3	8.7/3.9	-	-	583

4. CONCLUSIONS

The Tupiguarani groups occupied the northern coast of Pernambuco around 1200 BP, with similar chronologies across sites. Mineralogical and chemical composition determined by XRF shows low content of other minerals, high quartz content in silt and medium grains, and clays with significant Al₂O₃, suggesting intentional use of quartz as a temper and careful selection of raw materials to improve paste workability. Variations in Fe₂O₃ between sites may explain differences in color and EPR signals. Estimated firing temperatures (425–650 °C) are consistent with the chemical composition, indicating an uncontrolled firing process sufficient to confer functional strength to the vessels. Techniques such as OSL, EPR, XRD, and XRF allow reconstruction of these technological practices.

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